AI-Driven Dynamic Narrative Personalization in Games

MSc Computer Science / HCI

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Research Problem & Motivation

The Gap

- **Commercial practice:** One-size-fits-all narratives; broad difficulty options but static story
- **Prior AI systems:** Either siloed (player model *or* narrative generation) or static LLM prompts
- **Opportunity:** Closed-loop integration of telemetry → modeling → narrative → RL adaptation

> **Key Stat:** 42% of players cite narrative quality as critical to engagement, yet <5% of commercial games personalize story dynamically (Quantic Foundry, 2024)

Research Gap

Research Questions & Hypotheses

Research Question	Hypothesis
RQ1: Does real-time player modeling from telemetry improve narrative fit?	**H1:** GEQ Flow subscale scores significantly higher in experimental condition (dynamic) vs. control (static) *(primary)*
RQ2: Does narrative generation with episodic memory enhance immersion?	**H2:** GEQ Immersion subscale scores significantly higher in experimental group
RQ3: Can RL optimize narrative actions toward flow states?	**H3:** NPS-3 personalization perception significantly higher; PPO converges to >0.4 mean return

Study Design: Between-subjects, $N=50$ (25 experimental, 25 control), Welch's t-test, $\alpha=0.05$

System Architecture Overview

```
graph LR
    A["Unity Game<br/>(AnyRPG)"] -->|Telemetry<br/>WebSocket| B["Player Data<br/>Logger"]
    B -->|5-sec windows| C["SQLite<br/>Telemetry DB"]
    C -->|11 features| D["Player Modeling<br/>(Hexad + Flow)"]
    D -->|REST| E["Narrative Gen<br/>(RAG + Ollama)"]
    E -->|REST| F["Adaptation<br/>Engine<br/>(PPO MDP)"]
    F -->|NarrativeAction| G["Unity Plugin<br/>ContentInjector"]
    G -->|In-game update| A

    D -.->|HTTP fallback| E
    E -.->|Phi-3.5 Mini 3.8B| H["Ollama<br/>Fallback: Llama 3.2 3B"]

    style A fill:#e8f4fd
    style C fill:#fff4e6
    style H fill:#e8f5e9
```

Tech Stack: FastAPI 0.115+ | Python 3.14 | Pydantic v2 | Gymnasium | SB3 | Unity 6 LTS

Module 1: Player Data Logger

Telemetry Collection

- **23 fields** captured every **5 seconds** from Unity gameplay
- **Transport:** WebSocket primary (real-time), HTTP POST fallback (reliability)
- **Storage:** SQLite (persistent, query-friendly)

Field	Type	Example	Purpose
`timestamp`	ISO8601	2026-02-28T14:30:45Z	Session alignment
`position`	(x, y, z)	(-5.2, 1.0, 12.3)	Exploration metric
`velocity`	float	3.5	Engagement proxy
`challenge_level`	int	8	Difficulty context
`skill_level`	int	6	Player competence
`game_state`	enum	COMBAT, EXPLORATION	Context

Module 2: Player Modeling (Part A: Hexad Profile)

Hexad Continuous Profile from Telemetry

6-dimensional player type vector (not questionnaire) derived from 11 extracted gameplay features:

Feature	Contribution	Weight Formula
`combat_initiated`	Achiever	`(count_combats / total_actions) * 100`
`areas_explored`	Explorer	`(unique_areas / map_zones) * 100`
`npc_interactions`	Socializer	`(npc_talks / playtime_min)`
`quest_abandonment`	Free Spirit	`(abandoned_quests / total_quests)`
`rule_breaking`	Disruptor	`(OOB_detections + cheats) / session_length`
`npc_gifts_given`	Philanthropist	`(gifts_given / npc_count)`

Output: 6-dim continuous vector in [0, 100], normalized per dimension

Module 2: Player Modeling (Part B: Flow States)

Flow State Classifier (Rule-Based, 4 States)

```
stateDiagram-v2
    [*] --> IDLE: idle_duration > 40% <br/> AND disengagement_signal
    IDLE --> APATHY: idle_duration > 40% <br/> (continues)
    APATHY --> BOREDOM: player_active <br/> but challenge_skill_ratio < 0.75
    BOREDOM --> FLOW: ratio in [0.75, 1.30]
    FLOW --> ANXIETY: ratio > 1.30
    ANXIETY --> BOREDOM: ratio < 0.75
    FLOW --> APATHY: timeout > 5 min

    style APATHY fill:#d32f2f
    style BOREDOM fill:#f57c00
    style FLOW fill:#388e3c
    style ANXIETY fill:#1976d2
```

Key Metric: `challenge_skill_ratio = challenge_level / skill_level`

State	Condition	Reward	Color
FLOW	0.75 ≤ ratio ≤ 1.30	+1.0	Green

Module 3: Narrative Generation Pipeline

```
sequenceDiagram
    participant PM as Player Model
    participant RAG as RAG Retriever
    participant EM as Episodic Memory
    participant PB as Prompt Builder
    participant LLM as Ollama (Phi-3.5)
    participant NC as NarrativeContent

    PM->>RAG: Query: player_state + context
    RAG->>RAG: Embed query (all-MiniLM-L6-v2<br/>384-dim)
    RAG->>RAG: Cosine similarity search<br/>top-3 lore snippets
    EM->>EM: Select top-5 weighted<br/>episodic events (max 10)
    RAG->>PB: Lore context
    EM->>PB: Session memory
    PM->>PB: Hexad profile + flow state
    PB->>PB: Build dynamic prompt<br/>(1800 tokens)
    PB->>LLM: Prompt + JSON schema
    LLM->>LLM: Generate narrative action<br/>(8 options)
    LLM->>NC: NarrativeAction + reason
    NC-->>PM: Update game state
```

Tech: sentence-transformers all-MiniLM-L6-v2 | Cosine similarity | Fallback: Llama 3.2 3B

Module 4: Adaptation Engine (PPO + MDP)

Markov Decision Process Formulation

Component	Definition	Size
State Space	[hexad_6, flow_state_1, challenge_5, duration_2, context_embed_1]	7-dim continuous
Action Space	{LOWER_STAKES, RAISE_STAKES, ADD_MYSTERY, ADD_HUMOR, PROVIDE_GUIDANCE, INCREASE_URGENCY, LORE_REWARD, NO_CHANGE}	Discrete(8)
Reward	R(s,a) = flow_bonus + state_penalty + exploration_bonus	Per transition

Reward Function Breakdown

```
R(s, a) =
  +1.0  if new_flow_state == FLOW
  -0.5  if new_flow_state == ANXIETY
  -0.3  if new_flow_state == BOREDOM
  -0.8  if new_flow_state == APATHY
  +0.1  × (hexad_explorer_score / 100)  [curiosity_bonus]
```

Module 5: Unity Plugin (4 C# Components)

Component Roles

Component	Responsibility	Key Method
PluginConfig	ScriptableObject; backend URL, auth token, update frequency	`GetInstance()`
PlayerDataLogger	MonoBehaviour; collects 23 telemetry fields, sends via WebSocket/HTTP	`LogPlayerState()`
NarrativeManager	REST client; calls `/api/narrative/generate`, caches actions	`RequestNarrativeAction()`
ContentInjector	Applies NarrativeActions to game state (AnyRPG hooks)	`ApplyAction(NarrativeAction)`

Developer Setup (3 Steps)

Implementation Progress

Category	Task	Status	Notes
Backend	FastAPI scaffold + Pydantic models	✅ Done	Config-driven
Backend	SQLite telemetry storage	✅ Done	Indexed on player_id, timestamp
Backend	WebSocket + HTTP telemetry endpoints	✅ Done	Dual transport
Backend	Feature extractor (11 features)	✅ Done	Tested on 100K events
Backend	Flow state classifier	✅ Done	4 states, rule-based
Backend	Hexad profiler (6-dim continuous)	✅ Done	Weights validated
[illegible]	[illegible]	✅	[illegible]

Implementation Progress (cont.)

Category	Task	Status	Notes
RL Engine	Gymnasium RL environment (MDP)	✅ Done	7-dim obs, Discrete(8) actions
RL Engine	Reward function	✅ Done	Flow-based, +exploration bonus
RL Engine	PPO agent (cold-start + lazy)	✅ Done	SB3, ready for 200k training
RL Engine	Narrative generation REST endpoint	✅ Done	JSON mode, <500ms p95
Unity	PluginConfig ScriptableObject	✅ Done	Inspector-editable
Unity	PlayerDataLogger MonoBehaviour	✅ Done	23 fields, WebSocket/HTTP
Unity	NarrativeManager REST client	✅	Async queue, retry logic

Implementation Progress: Remaining

Category	Task	Status	Target
Integration	Testbed game (AnyRPG Core hookup)	⏳ In Progress	Week 2, Mar 2026
RL	PPO training run (200k timesteps)	⏳ Pending	Week 3, Mar 2026
Study	User study (N=50, between-subjects)	⏳ Pending	Apr–May 2026
Study	Data collection & analysis	⏳ Pending	May–Jun 2026
Thesis	Thesis write-up completion	⏳ Pending	Jun–Jul 2026

Test Suite Badge: 59/59 tests passing (Backend: 35 | RL: 15 | RAG: 9)

Novelty Contributions

Comparison of this work vs. prior systems on 6 key dimensions:

Dimension	This Work	PANGeA	LIGS	Traditional DDA
Player Profile	Hexad continuous (telemetry)	Big Five (fixed)	None	Difficulty slider
Narrative Gen	RAG + LLM (grounded)	Static prompts	Emergent (no grounding)	N/A
Session Memory	Episodic (top-5 events injected)	None	None	N/A
Action Selection	PPO (RL)	Heuristic rules	Fixed sampling	Ad-hoc
Flow Observable	challenge_skill_ratio MDP	Not modeled	Not modeled	Assumed
Integration	Full closed-loop (5 modules)	Partial (2 modules)	Partial (1 module)	Decoupled

Challenges & Solutions

Challenge 1: Finding a Suitable Open-Source Testbed Game

Problem: Needed a free, open-source Unity game with NPC dialogue, combat, quests, and exploration to serve as a realistic research testbed — building one from scratch was out of scope.

Solution: Selected **AnyRPG Core** (github.com/AnyRPG/AnyRPGCore)

- 100% free and open-source Unity RPG engine (MIT licence)
- Built-in NPC dialogue, quest system, combat, lore pickups, and scene transitions
- All gameplay events map directly to `PlayerDataLogger` public API hooks
- Compatible with Unity 6 LTS

Challenge 2: Cold-Start RL Dilemma

Problem: PPO requires interaction data to train, but no policy data exists before first deployment — the agent cannot make meaningful decisions in early sessions.

Solution: Heuristic bootstrap for the first **120 seconds** of each session:

Flow State	Heuristic Action
FLOW	NO_CHANGE
BOREDOM	INCREASE_URGENCY
ANXIETY	PROVIDE_GUIDANCE
APATHY	ADD_MYSTERY

After 120s, PPO takes over with accumulated session transitions.

Challenge 3: Hardware Limitations

Problem: Running a local LLM (Phi-3.5 Mini 3.8B) alongside PPO training and the FastAPI backend simultaneously places significant demand on consumer hardware — high latency risks breaking real-time immersion during the user study.

Solution: Staged execution strategy:

- PPO trained **offline** (200k timesteps) before the study begins; inference-only during sessions
- Ollama runs Phi-3.5 Mini in **4-bit quantised** mode — reduces VRAM from ~8GB to ~3GB
- Narrative requests fire **asynchronously** every 30s — not on every player action
- Target latency: full pipeline (telemetry → narrative) **< 3 seconds** on study hardware

Evaluation Design

Study Design Overview

```
Recruitment (N=50)
    ↓
Informed Consent & Baseline Demographics
    ↓
├─ Experimental (n=25): Dynamic Narrative (PPO + RAG)
└─ Control (n=25): Static Narrative (no adaptation)
    ↓
Play Session (45 min, counterbalanced quest)
    ↓
Post-Play Questionnaires (GEQ, miniPXI, NPS-3)
    ↓
Semi-Structured Interview (15 min, ~10 participants)
    ↓
Statistical Analysis (Welch t-test, RTA)
```

Randomization: Stratified by gaming experience (casual vs. core) to balance confounds

Instruments & Scales

Instrument	Items	Scale	Target Cronbach's α	Primary Hypothesis
GEQ (Game Experience Questionnaire)	4 (Flow subscale)	1–5 Likert	≥ 0.70	H1 (primary DV)
GEQ	4 (Immersion subscale)	1–5 Likert	≥ 0.70	H2
miniPXI (Player Experience Inventory)	5 (Competence)	1–5 Likert	≥ 0.65	Exploratory
NPS-3 (Narrative Personalization Scale)	3 (custom)	1–7 Likert	≥ 0.68	H3
Semi-Structured Interview	8–10 open-ended	Qualitative	N/A (RTA)	Emergent themes

Justification for N=50: Power analysis ($\alpha=0.05$, $1-\beta=0.80$, effect size $d \geq 0.6$) requires $n \geq 36$ per group; $n=25$ per group provides buffer for attrition + secondary metrics.

Analysis Plan

Quantitative Analysis

1. **Assumption Testing:**
    - Shapiro-Wilk normality test on GEQ Flow subscale scores
    - Levene's test for homogeneity of variance
2. **Primary Hypothesis (H1):**
    - Welch's t-test (robust to unequal variance) on GEQ Flow subscale
    - H0: $\mu_{exp} = \mu_{ctrl}$ vs. H1: $\mu_{exp} > \mu_{ctrl}$ (one-tailed)
    - $\alpha=0.05$, report p-value and 95% CI
3. **Effect Size:**
    - Cohen's d (pooled SD) with interpretation: $|d| \geq 0.5$ = medium, $|d| \geq 0.8$ = large

Qualitative Analysis

Approach: Reflexive Thematic Analysis (Braun & Clarke, 2022)

Phases:

1. **Familiarization:** Transcribe 10 interviews; read multiple times
2. **Coding:** Inductive open codes; focus on player perceptions of narrative personalization
3. **Theme Development:** Group codes into candidate themes (e.g., "Perceived Agency", "Immersion Breaks", "Story Coherence")
4. **Review & Refinement:** Validate against full interview dataset; reflexivity journal
5. **Write-Up:** Illustrative quotes per theme; integrate with quantitative results

Integration: Triangulate qualitative insights with t-test outcomes to explain mechanisms of flow increase.

Timeline & Next Steps

```
gantt
    title Remaining Project Milestones
    dateFormat YYYY-MM-DD

    section Testbed
    AnyRPG Integration       :s1, 2026-02-28, 14d
    Play Session Pilot (n=3) :s2, after s1, 7d

    section RL Training
    PPO Training Run (200k)  :s3, after s2, 10d
    Policy Evaluation        :s4, after s3, 5d

    section User Study
    Participant Recruitment  :s5, 2026-03-25, 21d
    Main Study Sessions      :s6, after s5, 30d

    section Analysis
    Data Processing & Checks :s7, after s6, 10d
    Statistical Tests        :s8, after s7, 10d
    Qualitative Analysis     :s9, after s8, 14d

    section Thesis
    Results Write-Up         :s10, after s9, 14d
    Full Thesis Draft        :s11, after s10, 21d
    Final Submission         :s12, after s11, 3d
```

Demo & System Running

Live Endpoints (Available Now)

```
GET  http://localhost:8000/health
     └─ Response: {"status": "ok", "timestamp": "2026-02-28T..."}

POST http://localhost:8000/api/telemetry
     └─ Payload: {"player_id": "p_001", "position": [...], "challenge_level": 8}

GET  http://localhost:8000/api/player-model/p_001
     └─ Response: {
          "hexad": {"achiever": 75, "explorer": 60, ...},
          "flow_state": "FLOW",
          "timestamp": "2026-02-28T14:30:45Z"
        }

POST http://localhost:8000/api/narrative/generate
     └─ Payload: {"player_id": "p_001", "context": "combat"}
     └─ Response: {
          "action": "ADD_MYSTERY",
          "narrative_text": "A strange shadow flickers...",
          "confidence": 0.87
        }
```

Questions & Summary

Key Takeaways

- **Novel approach:** Closed-loop integration of telemetry → Hexad modeling → RAG narrative → PPO adaptation (6 dimensions, first-of-its-kind)
- **Implementation:** 59/59 tests passing; all 5 modules functional; FastAPI backend + Unity plugin production-ready
- **Evaluation:** Between-subjects RCT (N=50) with validated instruments (GEQ, miniPXI, NPS-3) + qualitative RTA

Next Immediate Steps

1. **This week (Feb 28 – Mar 7):** Complete AnyRPG testbed integration + pilot (n=3)
2. **Mar 8 – 18:** Run PPO training (200k timesteps); evaluate policy convergence

Thank You

Research Vision: Demonstrating that closed-loop AI narrative systems can significantly increase player Flow and immersion through continuous modeling and RL-driven adaptation.

Questions?

Contact: [Email] | Institution: [University Name]